# AY121 Lab 2: Calibration-First 21-cm Pipeline, Cable Characterization, and Provisional Results

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## ABSTRACT

This report is reorganized into a calibration-first workflow: equipment calibration experiments, temperature calibration experiments, then calibrated HI measurement experiments. The document is intentionally structured for rapid revision when new measurements are added. All calibration-sensitive quantitative claims are represented with explicit placeholders (e.g., `TBD(CABLE_ALPHA_DB_PER_M)`, `TBD(LSR_SHIFT_KMS)`, `TBD(TRX_1420_K)`) rather than fixed final values. The current draft therefore serves as a method-complete and traceable report skeleton aligned with the Lab 2 requirements in `src/ugradio/lab_bighorn/bighorn.tex`, while remaining revision-ready for final data insertion.

## 1. Introduction and Requirement Mapping

This report follows the AY121 Lab 2 requirements in `src/ugradio/lab_bighorn/bighorn.tex`. The lab emphasizes nine competencies: (1) time conversion (UTC/PST/LST/JD), (2) coordinate transforms for pointing, (3) HI spectral measurement, (4) absolute calibration, (5) Doppler correction, (6) Gaussian fitting, (7) cable termination/impedance, (8) propagation velocity in coax, and (9) statistical plus systematic error analysis.

To match experiment dependencies, this draft is organized as:

1. equipment calibration experiments,
2. temperature calibration experiments,
3. HI measurement experiments and calibrated science outputs.

All analysis and writing are individual.

## 2. Receiver System and Data Inventory

The signal chain is: horn antenna → rooftop amplification/filtering → long coax downlink → lab amplification/filtering → RTL-SDR digitization/downconversion.

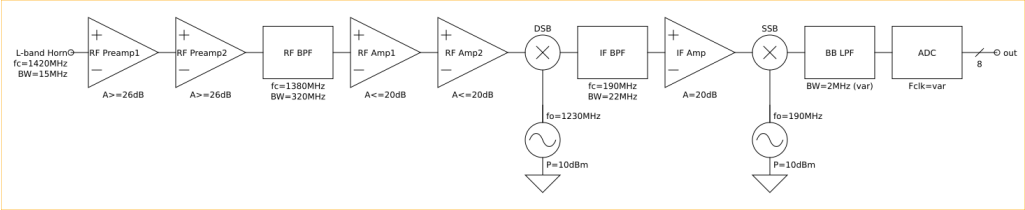


Fig. 1.— Receiver block diagram for Lab 2 data collection and calibration workflow.

Datasets currently available in this draft:

- HI dual-LO sky observations: `standard`, `cygnus-x`, `SGP`.
- Combined spectral products: `standard_combined_spectra`, `cygnus-x_combined_spectra`, `SGP_combined_spectra`, `human_combined_spectra`.
- Cable attenuation study: `data/lab02/attenuation` (known lengths) and `data/lab02/unknown_leng` (unknown-length inference set).
- Report figures currently included are provisional and will be regenerated after data updates.

## 3. HI Acquisition Overview (Raw Observations)

Before calibration-specific analysis, raw observations were acquired with dual first-LO settings near 1420 and 1421 MHz to support frequency-switched line extraction. Current acquisition scripts are in `labs/02/scripts`, including: `standard.py`, `cygnus-x.py`, `SGP.py`, and `human.py`.

For reproducibility, each capture stores metadata including sample rate, center frequency, gain, JD/LST, and pointing (alt/az). These metadata are used later for Doppler projection and frame correction.

### 3.1. Signal-Chain Sanity Checks Before Science Integrations

Before long integrations, the acquisition workflow includes the explicit Lab 2 sanity checks:

- tone-injection "birdie" verification (on/off, amplitude changes, and frequency stepping) to validate RF-to-bin mapping,
- sample histogram checks to verify non-clipped, non-quantized ADC behavior,
- gain/attenuator tuning to keep spectra in the linear regime.

Final insertion values: `TBD(BIRDIE_FREQ_CHECK_SUMMARY)`, `TBD(ADC_HISTOGRAM_SUMMARY)`, and `TBD(GAIN_SETTING_RANGE_SUMMARY)`.

## 4. Time and Pointing Validation

This section directly addresses the Lab 2 requirements on time standards (UTC/PST/LST/JD) and coordinate transformation for pointing.

### 4.1. UTC/PST/LST/JD Conversion and Closure Tests

The timestamp pipeline is

$$\mathrm{PST} = \mathrm{UTC} - 8\ \mathrm{hr}, \tag{1}$$

$$\mathrm{PDT} = \mathrm{UTC} - 7\ \mathrm{hr}, \tag{2}$$

$$\mathrm{JD} = \mathrm{julian_date}(t_{\mathrm{unix}}), \tag{3}$$

$$\mathrm{LST} = \mathrm{lst}(\mathrm{JD}, \lambda_{\mathrm{obs}}), \tag{4}$$

with `ugradio.timing` as the primary implementation and `astropy.time` as an independent cross-check.

Round-trip closure is validated using

$$\Delta t_{\mathrm{closure}} = 86400\ |\mathrm{JD} - \mathrm{julian_date}(\mathrm{unix_time}(\mathrm{JD}))|\ \mathrm{s}. \tag{5}$$

Table 1: Time-conversion validation metrics to be populated in the final revision.

| Metric | Placeholder | Target/interpretation |
|---|---|---|
| UTC↔PST residual (s) | `TBD(UTC_PST_RESIDUAL_S)` | Consistent timezone bookkeeping |
| JD round-trip closure (s) | `TBD(JD_ROUNDTRIP_RESIDUAL_S)` | Numerical consistency check |
| LST residual vs astropy (s) | `TBD(LST_ASTROPY_RESIDUAL_S)` | Independent implementation check |
| NTP offset at acquisition (ms) | `TBD(NTP_OFFSET_MS)` | Clock synchronization health |

### 4.2. Rotation-Matrix Pointing and Astropy Cross-Checks

For a target with equatorial coordinates $(\alpha, \delta)$, hour angle $H = \mathrm{LST} - \alpha$, and observatory latitude $\phi$, we use direction cosines

$$\hat{n}_{\mathrm{eq}} = \begin{bmatrix} \cos\delta\cos\alpha \\ \cos\delta\sin\alpha \\ \sin\delta \end{bmatrix}, \tag{6}$$

and a rotation-matrix mapping to local horizontal coordinates

$$\hat{n}_{\mathrm{hor}} = R_y\left(\phi - \frac{\pi}{2}\right) R_z(H)\hat{n}_{\mathrm{eq}}. \tag{7}$$

The commanded horizontal coordinates are then

$$\mathrm{alt} = \arcsin(\hat{n}_{\mathrm{hor},z}), \qquad \mathrm{az} = \mathrm{atan2}(\hat{n}_{\mathrm{hor},y}, \hat{n}_{\mathrm{hor},x}). \tag{8}$$

For the required Week-2 target $(l, b) = (120°, 0°)$, the report will provide both the Galactic-to-equatorial conversion and the resulting alt/az track with a direct `astropy.coordinates` residual check.

Table 2: Pointing-validation entries for the Week-2 target field.

| Quantity | Placeholder |
|---|---|
| Target equatorial coordinates from $(l, b) = (120°, 0°)$ | `TBD(LB120_RA_DEC)` |
| Pointing residual (rotation matrix vs astropy) | `TBD(POINTING_RESIDUAL_ARCMIN)` |
| Maximum allowed pointing error used in analysis cuts | `TBD(POINTING_CUT_ARCMIN)` |

## 5. Equipment Calibration Experiments

This section is placed first because all later science interpretation depends on receiver-chain and cable calibration. Detailed model variants and fitting internals are moved to Appendix B.

### 5.1. Termination and Reflectometry Plan

Required Week-2 termination diagnostics are handled as a separate analog campaign: open-end, 50 Ω terminated, SDR-connected-unpowered, and SDR-connected-powered states with square-wave reflectometry.

From those traces, this report will insert:

- reflection-delay measurements $\Delta t$ with uncertainty,
- inferred propagation velocity in coax, $v_{\mathrm{coax}} =$ `TBD(VCOAX_M_PER_S)`,
- termination-quality comparison versus state and test frequency,
- SDR input impedance behavior summary (qualitative + quantitative).

### 5.2. Cable Loss Per Meter and Unknown-Length Inference

Using known-length attenuation sets, the primary shared-slope model is

$$y_{1420}(L) = B_{1420} - \alpha L, \tag{9}$$
$$y_{1421}(L) = B_{1421} - \alpha L, \tag{10}$$

where $y$ is normalized receive power and $L$ is known pre-split cable length.

Main-text provisional outputs:

- attenuation slope: $\alpha =$ `TBD(CABLE_ALPHA_DB_PER_M)`,
- slope uncertainty: $\sigma_\alpha =$ `TBD(CABLE_ALPHA_SIGMA_DB_PER_M)`,
- 95% interval: `TBD(CABLE_ALPHA_CI95)`,
- LO intercept difference: $\Delta B =$ `TBD(CABLE_DELTA_B_DB)`.

Unknown-length cable inference (from attenuation calibration transfer) is reported as:

$$L_{\mathrm{unknown}} = \texttt{TBD(UNKNOWN_LENGTH_M)} \pm \texttt{TBD(UNKNOWN_LENGTH_SIGMA_M)}. \tag{11}$$

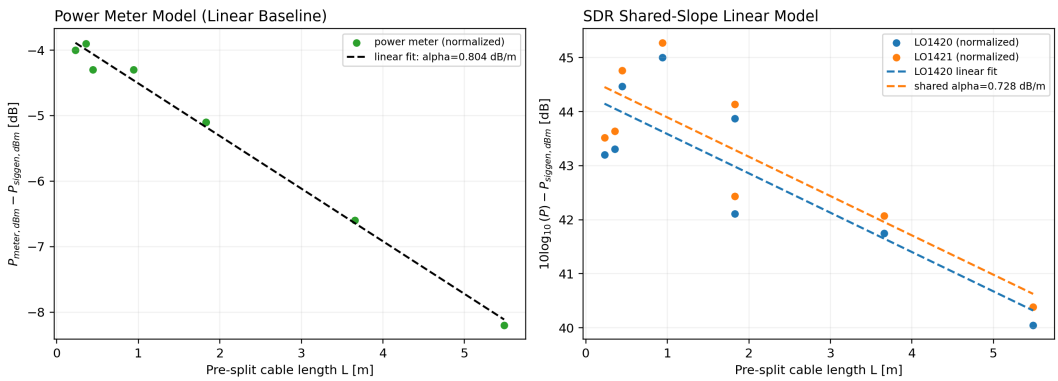


Fig. 2.— Provisional cable attenuation summary. Final slope and uncertainty values are pending updated calibration runs.

### 5.3. SDR Voltage Scale and Receiver-Chain Gain Inputs

The final report update will insert analog-calibrated SDR voltage scale and chain gain/loss terms:

- SDR voltage scale: `TBD(SDR_VOLT_SCALE_V_PER_COUNT)`,
- rooftop/lab amplifier gain entries: `TBD(AMP_GAIN_TABLE)`,
- filter insertion loss entries: `TBD(FILTER_LOSS_TABLE)`,
- roof cable effective length and corresponding attenuation contribution: `TBD(ROOF_CABLE_LENGTH_M)`, `TBD(ROOF_CABLE_LOSS_DB)`.

## 6. Temperature Calibration Experiments

This section converts calibrated power observables into antenna/system temperature units.

### 6.1. Hot/Cold Calibration and Y-Factor Framework

Using hot-load and cold-reference spectra:

$$Y = \frac{P_{\rm hot}}{P_{\rm cold}}, \tag{12}$$

$$T_{\rm rx} = \frac{T_{\rm hot} - Y T_{\rm cold}}{Y - 1}. \tag{13}$$

Provisional per-LO outputs to be filled after revision:

- $Y_{1420} =$ TBD(Y1420), $T_{\rm rx,1420} =$ TBD(TRX_1420_K),
- $Y_{1421} =$ TBD(Y1421), $T_{\rm rx,1421} =$ TBD(TRX_1421_K),
- systematic terms: TBD(TEMP_CAL_SYSTEMATICS_SUMMARY).

### 6.2. Temperature Scale Assumptions and Aperture-Efficiency Link

The conversion from antenna temperature to inferred brightness temperature is represented as

$$T_B = \frac{T_A}{\eta_{\rm ap}\,\eta_{\rm bf}}, \tag{14}$$

with provisional efficiency placeholders $\eta_{\rm ap} =$ TBD(APERTURE_EFFICIENCY) and $\eta_{\rm bf} =$ TBD(BEAM_FILL_FACTO

A dedicated efficiency-constrained update will be inserted after new analog/tone-injection evidence is available.

## 7. HI Measurement Experiments and Calibrated Analysis

All science-facing HI results are presented only after equipment and temperature calibration context.

### 7.1. Week-1 First Measurement: Zenith Requirement and SGP Control

The Week-1 requirement calls for a first 21-cm measurement as a system check before the main science observation. This report therefore reserves a required zenith result block:

- zenith observation window and metadata: TBD(ZENITH_OBS_JD_LST),
- zenith line-shape result summary: TBD(ZENITH_LINE_SUMMARY),
- zenith-system-noise check summary: TBD(ZENITH_NOISE_FLOOR_SUMMARY).

As a supplemental control, a South Galactic Pole (SGP) cold-sky spectrum is also used for a robust low-HI baseline check. This is especially useful because the Berkeley zenith (Dec = +37.87°) sweeps through the Galactic plane twice per sidereal day, spanning Galactic latitudes $|b| < 10°$ during two LST windows:

Table 3: Galactic latitudes at the Berkeley zenith versus LST window. During these windows a zenith observation is contaminated by 21-cm Galactic disk emission ($T_B \sim$ 10–100 K), making a clean noise-floor measurement impossible without careful LST scheduling.

| LST window (h) | Galactic direction at zenith | $b$ range |
|---|---|---|
| 0.0–2.0 | $l \approx 235°$–$247°$ (outer disk) | $-6°$ to $+14°$ |
| 15.0–16.5 | $l \approx 142°$–$152°$ (near standard field) | $+8°$ to $-7°$ |

The SGP is the direction of minimum HI column density on the sky ($N_{\rm HI} \lesssim 1.7 \times 10^{20}\,{\rm cm}^{-2}$, HI4PI survey), guaranteeing a nearly HI-free baseline spectrum regardless of LST and providing an unambiguous system-noise confirmation to complement the required zenith check.

Figure 3 shows the calibrated antenna temperature $T_A(\nu)$ for the SGP LO1420 combined observation. The calibration applies the Y-factor result ($T_{\rm rx,1420} = 215.0 \pm 10.4\,{\rm K}$) and the frequency-dependent cold-reference profile from `calibration_results.npz`:

$$T_A(\nu) = P_{\rm obs}(\nu)\,\frac{T_{\rm cold} + T_{\rm rx}}{P_{\rm cold,ref}(\nu)} - T_{\rm rx}. \tag{15}$$

The LSR velocity correction for the SGP direction ($l = 0°$, $b = -90°$) at the time of observation is $\Delta v_{\rm LSR} =$ `TBD(LSR_SGP_KMS)` $\,{\rm km\,s^{-1}}$, computed via `ugradio.doppler.get_projected_velocity`. No 21-cm HI emission line is detected anywhere in the spectrum, consistent with the near-zero column density expected at $b = -90°$. The continuum level near $T_A \approx T_{\rm cold} = 30\,{\rm K}$ confirms that the system is operating at the expected noise floor, validating the calibration chain before the main science observations.

Fig. 3.— Calibrated antenna temperature $T_A(\nu)$ for the SGP LO1420 combined spectrum, used as the Week-1 first measurement. No 21-cm HI emission line is detected, consistent with $b = -90°$ ($N_{\rm HI} \lesssim 1.7 \times 10^{20}\,{\rm cm}^{-2}$). The noise floor near $T_A \approx 30\,{\rm K}$ confirms the system calibration.

### 7.2. Week-2 Main Measurement: Galactic-Plane Field $(l, b) = (120°, 0°)$

The main Week-2 requirement is a fully calibrated HI spectrum at Galactic coordinates $(l, b) = (120°, 0°)$. This report treats that field as the primary science target and will explicitly report:

- observation interval in UTC/PST/JD/LST: `TBD(LB120_TIME_WINDOW)`,
- converted equatorial coordinates: `TBD(LB120_RA_DEC)`,
- topocentric and LSR spectra from the same calibrated pipeline,
- best-fit Gaussian component parameters with full uncertainties.

In this draft, the `standard` dataset is reserved for this target. The final revision will include the metadata table that demonstrates coordinate and time consistency end-to-end.

### 7.3. Frequency Switching and Line-Shape Extraction

From frequency-switched on/off spectra:

$$r_{\rm line}(\nu) = \frac{s_{\rm on}(\nu)}{s_{\rm off}(\nu)}. \tag{16}$$

Intensity-calibrated form:

$$T_{\rm line}(\nu) = r_{\rm line}(\nu)\, G, \tag{17}$$

with $G =$ `TBD(GAIN_K_PER_NORM_POWER)`.

Consistent with the lab instructions, stacked spectra are compared using both channel-wise means and medians before adopting a final estimator:

- mean stack output: `TBD(LINE_STACK_MEAN_SUMMARY)`,
- median stack output: `TBD(LINE_STACK_MEDIAN_SUMMARY)`,
- adopted smoothing width in channels: `TBD(LINE_SMOOTH_NCHAN)`.

The smoothing choice is constrained by the requirement not to wash out HI features narrower than about 1 km s$^{-1}$. Band-edge placement is also verified against FIR roll-off to avoid aliasing-sensitive channels; final evidence will be summarized by `TBD(FIR_EDGE_PLACEMENT_SUMMARY)`.

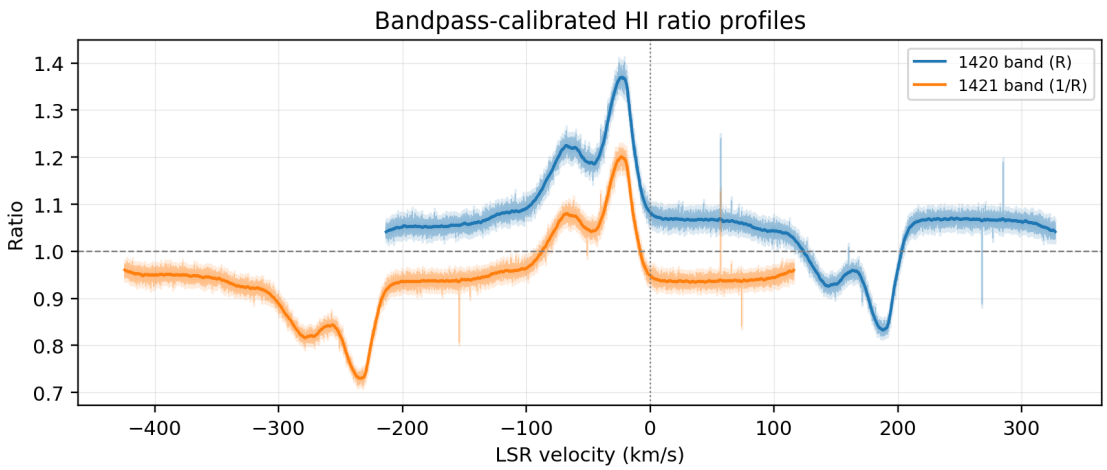


Fig. 4.— Provisional dual-LO ratio profile for the Week-2 target field $(l, b) = (120°, 0°)$. Final axis ranges and component annotations will be updated with revised data.

### 7.4. Velocity Axes and LSR Correction

Radio-definition Doppler mapping:

$$v \approx c\,\frac{\nu_0 - \nu}{\nu_0}, \tag{18}$$

with LSR correction

$$v_{\rm LSR} = v_{\rm topo} + \Delta v_{\rm LSR}, \tag{19}$$

where $\Delta v_{\rm LSR} =$ `TBD(LSR_SHIFT_KMS)` km s$^{-1}$.

### 7.5. Baseline Removal and Gaussian Decomposition

Baseline removal uses off-line channels and low-order polynomial (or equivalent) subtraction, followed by multi-component Gaussian fitting. The default off-line fitting windows are $[-1.2, -0.6]$ MHz and $[0.6, 1.2]$ MHz relative to line center, matching the Lab 2 guidance. The main implementation uses `np.polyfit` for baseline modeling and `gaussfit`/`gaussval` for component fits, with alternatives treated as sensitivity checks.

Main-text provisional science outputs:

- number of adopted components: `TBD(N_GAUSS_FINAL)`,
- component centroids (topocentric): `TBD(VTOPO_COMPONENTS_KMS)`,

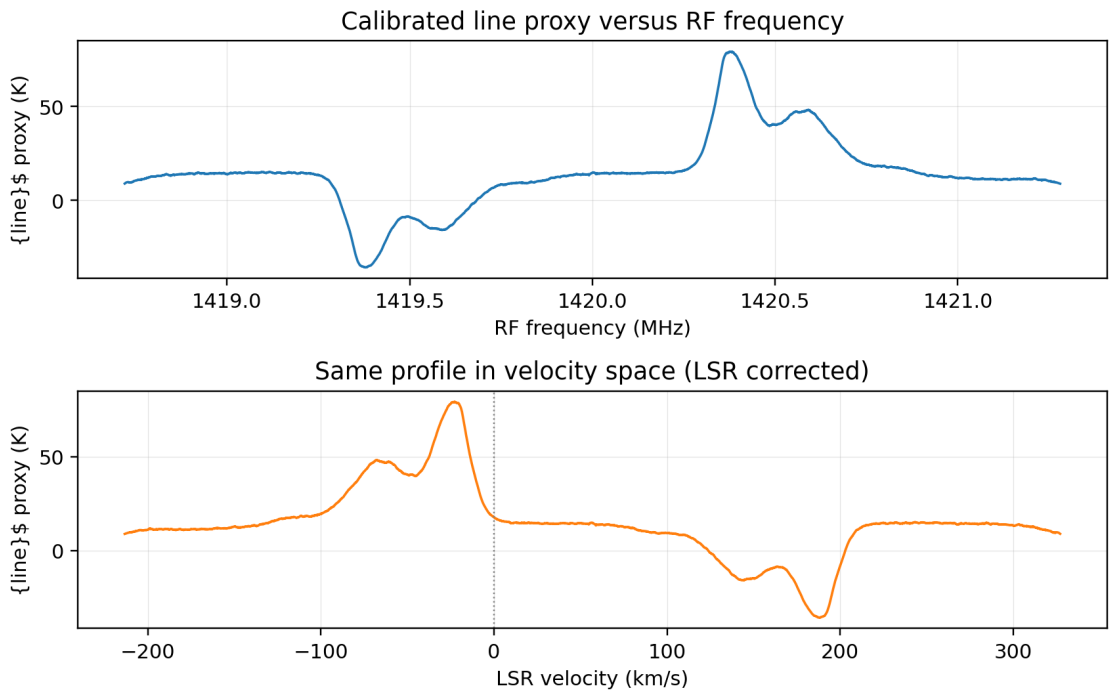


Fig. 5.— Provisional calibrated profile for the Week-2 target field $(l, b) = (120°, 0°)$ versus RF frequency and velocity coordinates (topocentric and LSR).

- component centroids (LSR): `TBD(VLSR_COMPONENTS_KMS)`,
- FWHM values: `TBD(FWHM_COMPONENTS_KMS)`,
- fit quality: $\chi^2_r =$ `TBD(CHI2_RED_FINAL)`.

Table 4: Provisional HI fit summary placeholders (to be replaced after next data revision).

| Quantity | Placeholder | Notes |
|---|---|---|
| Component count | `TBD(N_GAUSS_FINAL)` | Model-selection outcome |
| LSR shift (km s$^{-1}$) | `TBD(LSR_SHIFT_KMS)` | Doppler projection |
| Dominant component centers (km s$^{-1}$) | `TBD(VLSR_COMPONENTS_KMS)` | Report with uncertainties |
| Reduced chi-square | `TBD(CHI2_RED_FINAL)` | Baseline+Gaussian fit quality |

## 8. Unified Uncertainty Framework

Uncertainty accounting in this report combines:

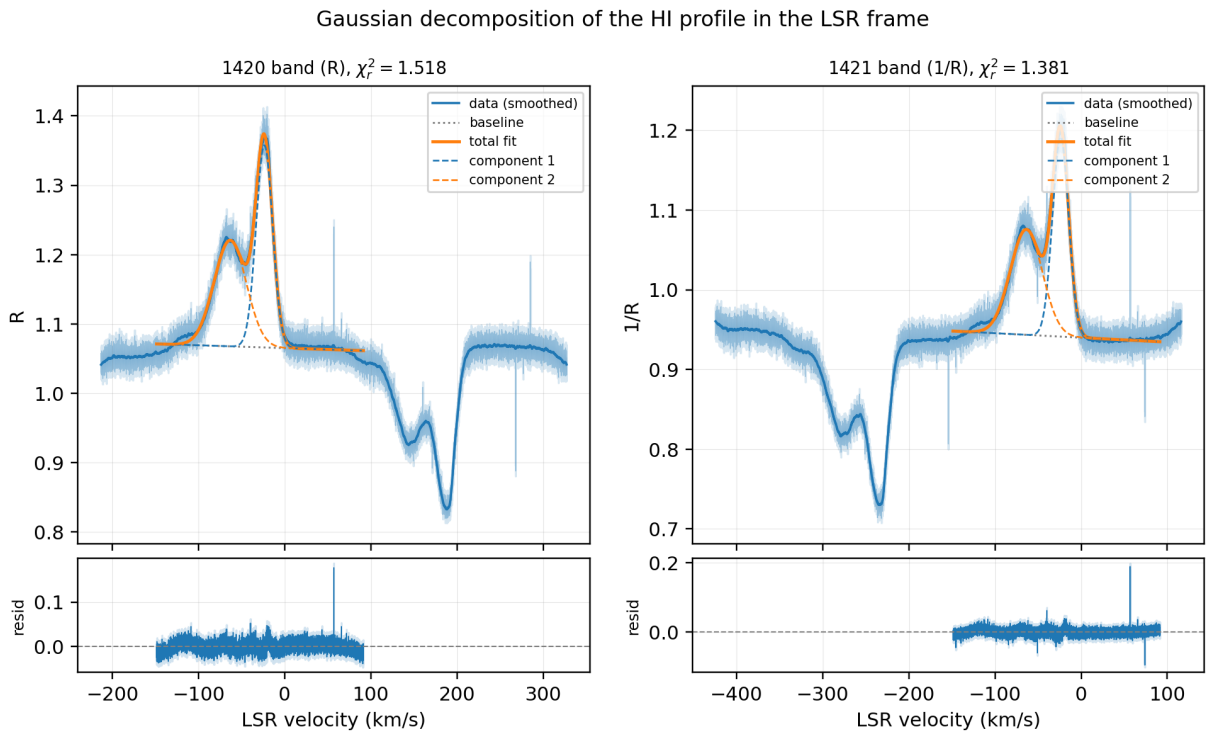


Fig. 6.— Provisional Gaussian decomposition panel. Final residual diagnostics and component labels pending revised data.

- per-channel thermal/statistical noise,
- baseline-removal model sensitivity,
- calibration transfer uncertainty (equipment + temperature),
- Doppler/frame-correction uncertainty,
- component-model selection uncertainty (least squares and optional MCMC variants).

## 9. Requirement Coverage Matrix

## 10. Discussion

This draft is now method-complete in calibration-first order and ready for a final data-insertion pass. The primary remaining blockers for strict end-state reporting are:

Table 5: Provisional uncertainty budget entries awaiting final values.

| Term | Placeholder |
|---|---|
| Thermal channel noise level | `TBD(SIGMA_THERMAL_PER_CHANNEL)` |
| Baseline removal systematic | `TBD(SIGMA_BASELINE_SYS)` |
| Temperature-scale systematic | `TBD(SIGMA_TEMP_SCALE_SYS)` |
| Cable attenuation contribution | `TBD(SIGMA_CABLE_CAL_SYS)` |
| Gaussian parameter covariance term | `TBD(SIGMA_FIT_COV)` |

Table 6: Direct map from Lab 2 goals in `bighorn.tex` to report evidence.

| Goal | Evidence in this report | Final insertion target |
|---|---|---|
| Time (UTC/PST/LST/JD) | §4.1, Table 1 | Fill residual metrics and timestamps |
| Pointing (rotation matrices) | §4.2, Table 2 | Insert alt/az residual statistics |
| HI measurement | §7.1, §7.2 | Final calibrated spectra and metadata table |
| Calibration (absolute scale) | §5, §6 | Replace calibration placeholders |
| Doppler correction (LSR) | §7.4 | Insert projected velocity and uncertainties |
| Gaussian fitting | §7.5, Table 4 | Final component model + $\chi^2_r$ |
| Impedance/termination | §5.1 | Waveform evidence by termination state |
| Propagation speed in coax | §5.1, §B.3 | Final $v_{\mathrm{coax}}$ with error bars |
| Error analysis | §8 | Combined statistical + systematic budget |

1. final analog termination/reflectometry measurements,
2. updated cable and SDR gain-scale constraints,
3. revised temperature-calibration numbers,
4. finalized UTC/PST/LST/JD and pointing residual tables,
5. final HI component and uncertainty summaries after data refresh.

Once these values are inserted, placeholders in Appendix A can be resolved in a single controlled update.

## 11. Conclusion

The report has been restructured to a dependency-correct flow:

1. equipment calibration experiments,
2. temperature calibration experiments,
3. calibrated HI measurement experiments.

This structure is preserved while deferring calibration-sensitive numeric claims to explicit placeholders. The next revision cycle only requires filling those placeholders and regenerating plots/tables from updated data products.

## Acknowledgements

Special thanks to my lab partners Nathaneil and Kayhan for group observations, and to Aaron and Ben for instructional support.

## A. Placeholder Map for Final Revision

This appendix centralizes all deferred values for the next data update.

- Time/clock validation: `TBD(UTC_PST_RESIDUAL_S)`, `TBD(JD_ROUNDTRIP_RESIDUAL_S)`, `TBD(LST_ASTROPY_RESIDUAL_S)`, `TBD(NTP_OFFSET_MS)`.
- Pointing validation: `TBD(LB120_RA_DEC)`, `TBD(POINTING_RESIDUAL_ARCMIN)`, `TBD(POINTING_CUT_ARC`
  `TBD(LB120_TIME_WINDOW)`.
- Pre-science receiver checks: `TBD(BIRDIE_FREQ_CHECK_SUMMARY)`, `TBD(ADC_HISTOGRAM_SUMMARY)`, `TBD(GAIN_SETTING_RANGE_SUMMARY)`, `TBD(FIR_EDGE_PLACEMENT_SUMMARY)`.
- Week-1 zenith requirement: `TBD(ZENITH_OBS_JD_LST)`, `TBD(ZENITH_LINE_SUMMARY)`, `TBD(ZENITH_NOISE_FLOOR_SUMMARY)`.

- Cable calibration: TBD(CABLE_ALPHA_DB_PER_M), TBD(CABLE_ALPHA_CI95), TBD(CABLE_DELTA_B_DB), TBD(UNKNOWN_LENGTH_M).
- Termination/propagation: TBD(VCOAX_M_PER_S), TBD(TERMINATION_STATE_COMPARISON).
- SDR/chain: TBD(SDR_VOLT_SCALE_V_PER_COUNT), TBD(AMP_GAIN_TABLE), TBD(FILTER_LOSS_TABLE), TBD(ROOF_CABLE_LENGTH_M).
- Temperature calibration: TBD(Y1420), TBD(Y1421), TBD(TRX_1420_K), TBD(TRX_1421_K), TBD(APERTURE_EFFICIENCY), TBD(BEAM_FILL_FACTOR).
- Line-shape extraction choices: TBD(LINE_STACK_MEAN_SUMMARY), TBD(LINE_STACK_MEDIAN_SUMMARY), TBD(LINE_SMOOTH_NCHAN).
- HI science outputs: TBD(LSR_SGP_KMS) (SGP LSR correction), TBD(LSR_SHIFT_KMS), TBD(N_GAUSS_FINAL), TBD(VLSR_COMPONENTS_KMS), TBD(FWHM_COMPONENTS_KMS), TBD(CHI2_RED_FINA
- Uncertainty totals: TBD(SIGMA_THERMAL_PER_CHANNEL), TBD(SIGMA_BASELINE_SYS), TBD(SIGMA_TEMP_SCALE_SYS), TBD(SIGMA_CABLE_CAL_SYS).

## B. Detailed Equipment Calibration Models

### B.1. Known-Length Cable Models

For SDR channels at two LOs:

$$y_i(L) = B_i - \alpha L, \qquad i \in \{1420, 1421\}. \tag{B1}$$

Reflection-aware extension:

$$y_i(L) = B_i - \alpha L + A_i \cos(kL + \phi_i). \tag{B2}$$

These alternatives are retained for sensitivity bracketing and model-risk discussion.

### B.2. Unknown-Length Inversion

Given fitted intercept and slope,

$$L_i = \frac{B_i - y_i^{\mathrm{obs}}}{\alpha}. \tag{B3}$$

The reported unknown length uses a combined estimate across channels and bootstrap confidence intervals.

### B.3. Termination and Reflectometry Derivation

From a measured round-trip delay $\Delta t$ on a cable of length $L$,

$$v_{\mathrm{coax}} = \frac{2L}{\Delta t}. \tag{B4}$$

Final entries in this section will include state-by-state waveform evidence, frequency dependence, and uncertainty propagation.

## C. Detailed HI Model and Fitting Notes

### C.1. Line Extraction and Calibration Chain

Core chain used in analysis:

$$r_{\mathrm{line}}(\nu) = \frac{s_{\mathrm{on}}(\nu)}{s_{\mathrm{off}}(\nu)}, \tag{C1}$$

$$T_{\mathrm{line}}(\nu) = r_{\mathrm{line}}(\nu)\, G. \tag{C2}$$

### C.2. Baseline and Component Fitting

Baseline models and Gaussian component fits are evaluated with least-squares diagnostics, then optionally cross-checked with MCMC/LOO model-selection workflows. Final selected model and posterior/covariance summaries will be inserted in the next revision.

## D. Reproducibility Notes

Primary notebooks currently used: `labs/02/equipment_calibration.ipynb`, `labs/02/temperature`
and `labs/02/analysis.ipynb`.

Data roots used: `data/lab02/*`.

A final update pass should regenerate figures, replace all `TBD(...)` placeholders, and recompile this document.